Category-Level Positive–Negative Prompting for Open-Set Cross-Scene Remote Sensing Image Classification

Xin Zhao, *Member, IEEE*, Yuanyuan Ye, Jun Lin, and Shengsheng Wang

***Abstract*—Cross-scene remote sensing (RS) image classification is crucial for urban analysis and environmental monitoring, but often suffers from distribution shifts caused by different imaging conditions. While unsupervised domain adaptation (UDA) can alleviate cross-scene discrepancies, its closed-set assumption is violated when unknown categories emerge in the target domain, thereby motivating the open-set domain adaptation (OSDA) setting. Existing OSDA methods remain limited in semantic modeling, since conventional visual approaches lack high-level semantic guidance and recent vision–language model (VLM)-based methods predominantly rely on positive prompts that encode only class-belonging semantics, providing limited cues for unknown-sample rejection. To address these issues, we propose category-level positive–negative prompting (CLPNP) for open-set cross-scene RS image classification. CLPNP integrates learnable positive prompts and multiple groups of negative prompts into a frozen VLM to jointly model belonging and nonbelonging semantics, thereby forming a category-level binary semantic discrimination mechanism. Built upon this prompt design, a positive–negative semantic-guided known–unknown separation (PNKUS) strategy is proposed, which establishes reliable binary semantic boundaries from labeled source data and transfers this discriminative structure to the target domain through uncertainty regularization, together with structured regularization on positive and negative prompts to ensure semantic diversity and separability. In addition, a threshold-free open-set inference mechanism is introduced by directly comparing positive and negative semantic responses for each known category. Extensive experiments on 18 cross-scene transfer tasks under two benchmark settings demonstrate that CLPNP consistently outperforms state-of-the-art OSDA methods and achieves a favorable balance between known-category recognition and unknown-category rejection.**

***Index Terms*—Cross-scene classification, open-set domain adaptation (OSDA), prompt learning, remote sensing (RS), vision–language model (VLM).**

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I. Introduction

CROSS-SCENE classification of remote sensing (RS) imagery plays an important role in applications, such as urban spatial structure analysis and dynamic water environment monitoring, and has attracted increasing research attention in recent years [1], [2], [3]. However, variations in imaging sensors, observation geometry, spatial resolution, and land cover characteristics introduce substantial distribution shifts across different scenes, which often lead to significant performance degradation under cross-scene deployment. Traditional classification models [4], [5], [6] are primarily based on supervised learning and heavily rely on large-scale, high-quality annotated data, resulting in limited generalization capability in cross-scene scenarios. To alleviate this issue, domain adaptation methods have been introduced and shown improved discriminative performance across scenes. Most existing approaches focus on the closed-set domain adaptation setting and attempt to reduce source–target distribution discrepancies through feature alignment or distribution matching [7], [8], [9]. However, these methods rely on the strong assumption that the source and target domains share an identical label space, which is often violated in real-world RS applications, where the target domain frequently contains unknown categories absent from the source.

To address this label-space mismatch, open-set domain adaptation (OSDA) extends the closed-set learning paradigm by jointly classifying shared known categories and rejecting unknown ones in the target domain [10]. Representative methods such as OSBP [11] impose open-set discrimination constraints in the classifier output space, while UADAL [12] aligns source and target known features and explicitly separates target unknown features via unknown-aware domain adversarial learning, avoiding negative transfer without threshold tuning. To further enhance unknown discrimination and feature alignment under open-set settings, several studies introduce multiclassifier architectures or auxiliary branch constraints. For example, STA [13] progressively separates samples using multiple classwise binary classifiers and a fine-grained discriminator guided by auxiliary constraints, whereas DAMC [14] adopts a multiclassifier adversarial framework that dynamically weights target samples according to their similarity to the source domain. In the RS field, several recent efforts have explored open-set cross-scene classification. SSOUDA [15] exploits contrastive self-supervised learn-

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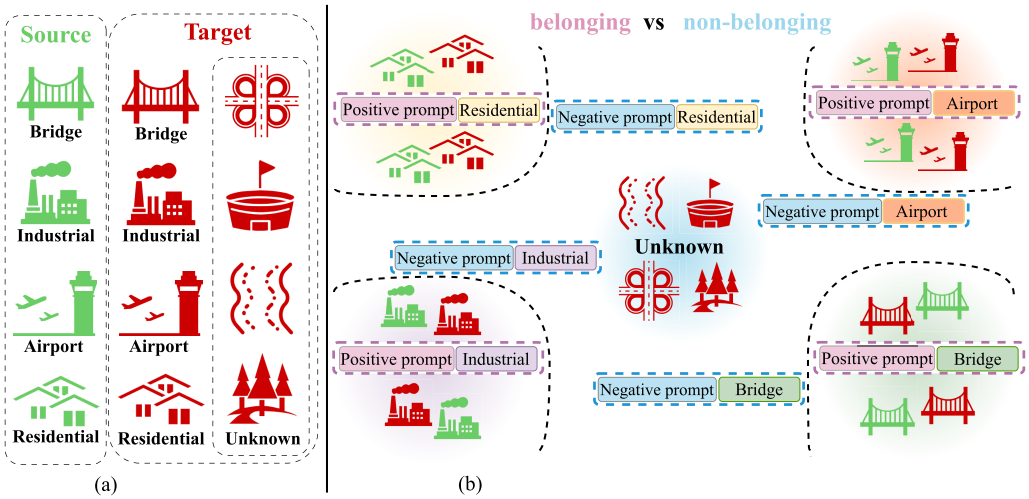


Fig. 1. Overview of the task setting and category-level positive–negative semantic mechanism of category-level positive–negative prompting (CLPNP). (a) Task setting of open-set cross-scene RS image classification. The source domain contains labeled samples from known categories, while the unlabeled target domain includes both shared known categories and unknown categories. (b) Through category-level positive–negative prompts, CLPNP establishes explicit category-level binary semantic discrimination between belonging and nonbelonging for each known class. Accordingly, rather than relying only on positive similarity matching of class-belonging semantics, the model explicitly compares the corresponding positive and negative semantic responses for each category, thereby enabling reliable known-class recognition and unknown-class separation.

ing together with consistency-based self-training to enable unknown-aware adaptation, while MAOSDAN [16] employs multibranch adversarial learning and attention-aware mechanisms to alleviate the interference of unknown samples during domain alignment. PUMCL [17] further proposes a prototype-based multiview consistency learning framework to enhance unknown-aware feature alignment and pseudo-label reliability. Despite these advances, most existing OSDA methods remain confined to purely visual, single-modality frameworks. Unknown discrimination is mainly achieved through auxiliary classifiers operating on visual feature embeddings, without high-level semantic guidance or cross-modal information fusion. As a result, under severe cross-scene distribution shifts, unknown samples are easily confused with known categories, and the diverse semantics of unknown categories cannot be effectively captured, which fundamentally limits the robustness of traditional OSDA methods in complex cross-scene RS scenarios.

To address the limitations of purely visual OSDA methods, recent studies have incorporated textual semantics to enhance open-set discrimination under cross-scene settings. By enabling cross-modal similarity learning between visual and textual features, vision–language models (VLMs) [18] combined with prompt learning have become a promising paradigm for OSDA [19], [20]. However, in practical OSDA scenarios, the absence of category names for unseen target classes prevents direct image–text alignment, making cross-domain semantic transfer challenging and motivating approaches that perform cross-modal alignment without explicit unknown labels. Representative methods include OSClip [21], which maps RS images and category-level textual descriptions into a shared vision–language embedding space and performs known–unknown discrimination using cosine similarity and energy scores under domain-adaptive prompt learning. COSMo [22], built upon CLIP [23], further explores prompt-based cross-modal alignment with frozen encoders and domain-adaptive components to enhance known–unknown separation. Despite the encouraging progress of existing VLM-based OSDA methods, they still suffer from clear limitations in prompt design and semantic modeling. Most of them rely solely on positive prompts for known categories, mainly emphasizing class-belonging semantics while lacking explicit modeling of class nonbelonging semantics. As a result, these methods often fail to establish complete and robust semantic decision boundaries for open-set recognition, especially under severe cross-scene distribution shifts. This shortcoming aligns with the broader finding that similarity-based matching alone is often insufficient in heterogeneous scenarios, while explicitly modeling nonmatching relationships provides complementary discriminative cues [24], [25]. To address this issue, we construct one group of positive prompts together with multiple groups of negative prompts for each known category (see Fig. 1), so that the model can jointly capture category-level binary semantic responses of belonging and nonbelonging. These prompts are implemented as learnable context vectors in the shared vision–language embedding space and are optimized to encode task-adaptive semantic context. In this way, the model explicitly compares belonging and nonbelonging semantic responses for each category, thereby establishing a category-level binary semantic discrimination relationship and laying the foundation for subsequent known–unknown separation in OSDA.

In summary, existing open-set cross-scene classification methods for RS still suffer from fundamental limitations in semantic modeling. Traditional unimodal approaches [see Fig. 2(a)] perform domain adaptation solely in the visual feature space, without leveraging textual semantics to provide high-level semantic guidance. Consequently, unknown samples are distinguished mainly according to low-level visual discrepancies, which are highly sensitive to cross-scene distri-

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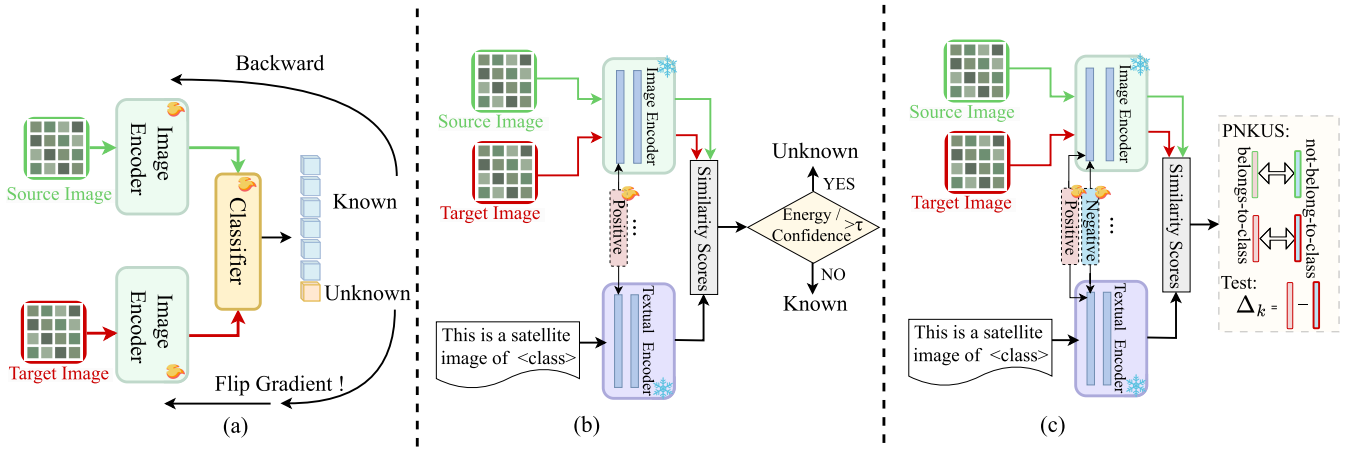


Fig. 2. Comparison of the semantic discrimination mechanisms between the proposed CLPNP and existing OSDA approaches. (a) Unimodal methods perform open-set recognition purely in the visual feature space, without semantic cross-modal guidance, which often leads to ambiguous known–unknown boundaries under cross-scene domain shifts. (b) Multimodal methods with only positive prompts enhance cross-modal matching for known categories, but still mainly rely on class-belonging semantics and lack explicit modeling of class nonbelonging semantics. (c) In contrast, CLPNP establishes a category-level binary semantic discrimination mechanism by jointly modeling belonging and nonbelonging semantics through positive and negative prompts, thereby enabling more reliable known–unknown separation and threshold-free open-set inference.

bution shifts and often result in ambiguous known–unknown decision boundaries. Recent VLM-based approaches [see Fig. 2(b)] incorporate textual information to improve feature transferability through cross-modal alignment between images and semantic descriptions. Although this design enhances generalization for known categories, these methods usually rely on a single positive prompt and therefore mainly model class-belonging semantics. In this case, unknown samples are mainly separated by their low responses to known-class prompts, rather than by explicit modeling of class nonbelonging semantics. As a result, the model lacks direct negative semantic cues for open-set discrimination, which weakens known–unknown separation under complex cross-domain shifts.

To address the above limitations, we propose CLPNP for open-set cross-scene RS image classification, as illustrated in Fig. 2(c). For each known category, CLPNP introduces one group of positive prompts and multiple groups of negative prompts, which are injected into a frozen vision–language encoder in a layerwise manner to explicitly construct a category-level binary semantic discrimination mechanism between belonging and nonbelonging. By jointly modeling positive and negative semantic responses for each category, CLPNP establishes more complete semantic decision boundaries than existing methods that rely solely on one-sided positive response, thereby enabling more robust known–unknown separation under cross-scene distribution shifts. Built upon this design, we further develop a positive–negative semantic-guided known–unknown separation (PNKUS) strategy for OSDA. In the source domain, PNKUS explicitly optimizes category-level binary semantic boundaries by enhancing the positive semantic response of the ground-truth category while enforcing negative-semantic dominance for the most confusing non-ground-truth category. In the target domain, PNKUS regularizes unlabeled target samples directly in the positive–negative semantic contrast space, encouraging clearer category-level binary semantic tendencies and more stable known–unknown separation under unsupervised conditions. Moreover, inference is performed by directly comparing positive and negative semantic responses, yielding a simple and threshold-free framework for cross-scene classification. The main contributions of this work are summarized as follows.

1) We propose CLPNP for open-set cross-scene RS image classification, a prompt-learning framework that jointly models class-belonging and class-nonbelonging semantics. By injecting one group of learnable positive prompts and multiple groups of negative prompts into a frozen vision–language encoder in a layerwise manner, CLPNP establishes category-level binary semantic representations and more complete semantic decision boundaries under cross-scene distribution shifts.
2) We develop a PNKUS strategy that unifies source-domain category-level semantic discrimination and target-domain uncertainty-regularized adaptation. Source-domain positive–negative semantic contrast sharpens category-level binary semantic boundaries, while target-domain uncertainty regularization reduces semantic ambiguity without relying on pseudo-labels. During inference, the relative semantic discrepancy between positive and negative semantic responses is exploited to realize a threshold-free open-set inference mechanism.
3) We introduce multiple groups of negative prompts together with two complementary structural regularization losses to enhance the diversity and discriminability of negative semantic representations while maintaining clear category-level separation between positive and negative semantics. This design prevents semantic collapse and strengthens open-set robustness under cross-scene variations.

II. Related Works

This section reviews related work on OSDA and prompt learning.

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A. Open-Set Domain Adaptation

Domain adaptation aims to mitigate cross-domain distribution shifts by transferring discriminative knowledge learned from labeled source-domain data to enable reliable classification of unlabeled target-domain samples. Early studies primarily focused on closed-set domain adaptation, which assumes that the source and target domains share an identical category space. Under this assumption, existing methods typically perform cross-domain adaptation by projecting source and target samples into a shared feature space and mitigating domain discrepancy via global domain alignment constraints, such as adversarial learning and distribution distance minimization, as well as category alignment strategies, including class center alignment and pseudo-label guidance [26], [27], [28], [29], [30]. However, in practical cross-scene RS scenarios, complex geographic environments and varying imaging conditions often introduce unknown categories in the target domain that are absent from the source domain, rendering the closed-set assumption invalid. To address this challenge, OSDA is formulated to preserve discriminative consistency for known categories while detecting and rejecting unknown samples in the target domain. The core challenge of OSDA lies in jointly modeling known-category alignment and unknown-category discrimination. STA [13] adopts a coarse-to-fine category separation strategy that integrates instance-level weighted adversarial adaptation and aligns only known-category distributions, thereby distinguishing known from unknown samples. ROS [31] employs a self-supervised rotation task to align source and known target distributions and separates known and unknown samples via rotation-based normality scoring.

Building upon these general OSDA formulations, recent studies have extended OSDA to cross-scene RS classification, where category space mismatch is further exacerbated by sensor diversity, scale variation, and complex scene semantics. Most RS-oriented OSDA methods primarily rely on visual feature learning to enhance the discriminability of known categories while improving the detection of unknown samples. OSDA-ETD [32] jointly optimizes transferability and discriminability by enforcing intraclass compactness and interclass separability, thereby enhancing discriminative robustness across domains. In terms of unknown category recognition, DFEN [33] adopts a parallel architecture that combines shared and private feature modeling, where the private branch captures discriminative cues specific to unknown target categories, improving unknown class detection. To address scenarios where the source model is inaccessible, OKRA [34] proposes an open-set black-box domain adaptation framework that enables joint known-category classification and unknown-category recognition without access to source model parameters. Although considerable progress has been made in OSDA, most existing methods still depend on purely visual feature learning and thus lack explicit semantic constraints for separating shared known classes from target-private unknown classes under domain shift. The emergence of VLMs offers a promising direction for RS OSDA by enabling cross-modal semantic transfer. Nevertheless, current prompt-based methods mainly rely on positive prompts to represent known-category semantics, without explicitly modeling class nonbelonging semantics that are essential for open-set discrimination. To address this issue, we introduce category-level binary semantic discrimination into prompt learning, so that both belonging and nonbelonging semantics are jointly encoded in the cross-modal space. Based on this design, we further develop a PNKUS strategy to achieve more reliable known–unknown separation in cross-scene RS OSDA.

B. Prompt Learning

Prompt learning is a parameter-efficient adaptation paradigm for pretrained CLIP and other VLMs. It leverages learnable prompt tokens to adapt pretrained knowledge to downstream tasks without fine-tuning the entire model, balancing adaptation performance and generalization. Early VLM prompt learning research primarily focused on textual prompt tuning, optimizing the text branch to generate task-aligned textual embeddings. CoOp [35] pioneers textual prompt learning by replacing fixed handcrafted templates with learnable context vectors, enabling automatic capture of task-relevant semantics. CoCoOp [36] extends this idea with image-conditioned prompt generation, which alleviates overfitting and enhances generalization. Recent advances have expanded prompt learning beyond the textual branch to multimodal and cross-modal scenarios. For example, MMA [37] deploys prompts at high-level layers of both visual and textual encoders based on dataset-level feature analysis. Modality-specific projection heads are introduced to learn task-specific knowledge, while a shared projection head is used to promote cross-modal alignment, thereby balancing discriminability and generalization. TCP [38] proposes a cross-modal class-aware prompt tuning strategy that embeds class-level textual knowledge into learnable prompts via textual knowledge embedding. These class-aware prompts are injected into intermediate layers of the text encoder and jointly optimized with domain-shared prompts using cross-modal contrastive and knowledge-guided consistency objectives.

In recent years, prompt learning has been increasingly introduced into OSDA. COSMo [22] adopts source-guided dual-branch prompts for known and unknown categories within a CLIP framework, incorporating domain bias via domain-specific batch normalization and pseudo-labeling with entropy regularization for multitarget adaptation. PromptDIV [39] exploits the domain invariance of CLIP textual features by combining one-versus-all clustering with text features (OVATs)-based unbiased pseudo-labeling with domain-specific prompts and mutual affinity contrastive learning to achieve domain-invariant alignment. Prompt learning has recently shown strong potential in RS applications, particularly for improving cross-scene robustness and open-set generalization through multimodal semantic adaptation. In the context of OSDA for RS, OSClip [21] performs dual-encoder prompt tuning with hierarchical language prompts and visual prompt accumulation, together with confidence-weighted pseudo-labeling and continual adaptation, enabling effective known–unknown recognition across multiple target domains. However, existing prompt-based RS OSDA methods mainly

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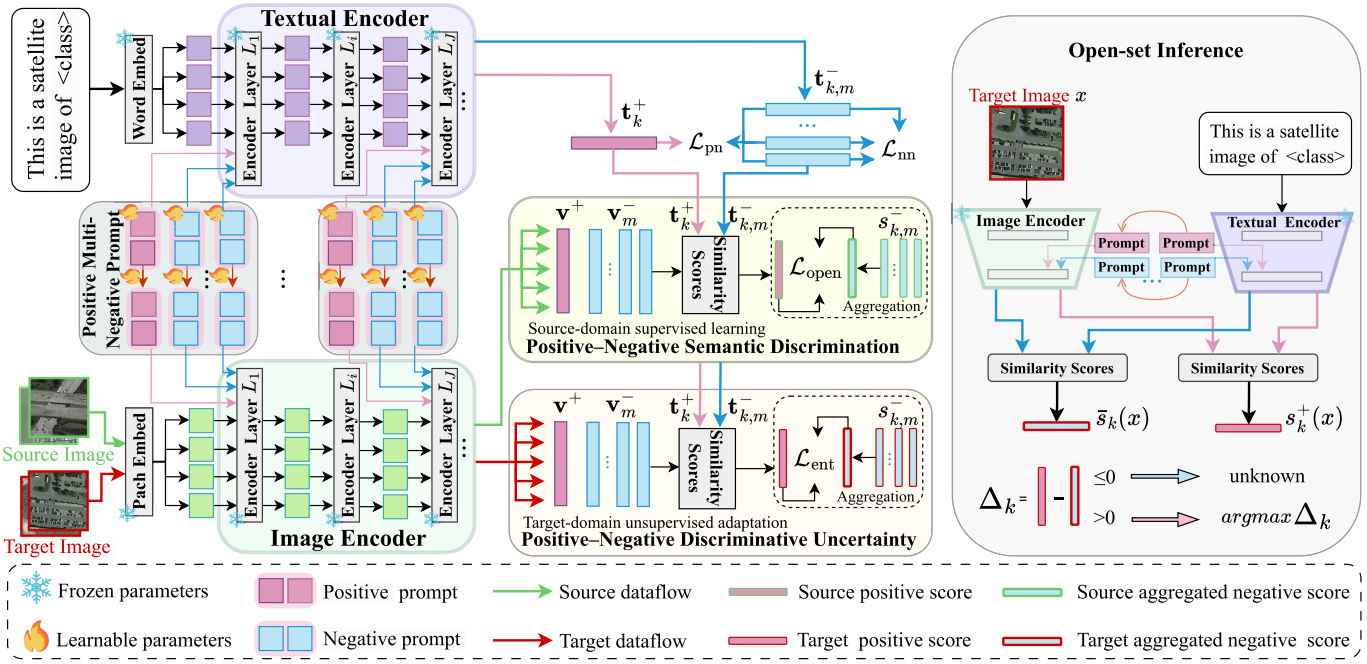


Fig. 3. Overview of the proposed CLPNP framework. CLPNP introduces a positive multinegative prompt module that injects learnable positive and negative prompts into a frozen vision–language encoder to construct category-level binary semantic representations. In the source domain, a positive–negative semantic discrimination module enforces category-level semantic contrast to strengthen decision boundaries between known categories and potential unknowns. In the target domain, a positive–negative discriminative uncertainty module suppresses prediction ambiguity for unlabeled samples. In addition, semantic diversity loss $\mathcal{L}_{nn}$ and discriminative margin loss $\mathcal{L}_{pn}$ are imposed on the prompt representations to ensure diversity among negative prompts and clear separation between positive and negative semantics. During inference, threshold-free open-set classification is performed based on the relative semantic discrepancy between positive and negative semantic scores.

rely on positive prompts to model known-category semantics, without explicitly modeling class nonbelonging semantics. This makes the model depend largely on known-class similarity, which is insufficient for robust known–unknown discrimination under cross-scene variations. To address this limitation, we introduce a structured prompt learning framework that jointly incorporates positive and negative prompts. By explicitly modeling category-level belonging and nonbelonging semantics within the prompt-induced semantic space, the framework establishes a more complete open-set semantic basis for known–unknown discrimination.

III. Proposed Method

This section presents the proposed CLPNP framework. An overview of the complete architecture is shown in Fig. 3.

A. Preliminaries

In practical RS applications, the source and target domains often exhibit significant distribution shifts due to variations in sensors, imaging viewpoints, and spatial resolutions. Under the OSDA setting, the target domain contains not only the known categories shared with the source domain but also samples from unknown categories that are absent from the source label space. Formally, the source domain is denoted as $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ with $y_i^s \in \mathcal{C}$, where $\mathcal{C} = \{1, 2, \ldots, C\}$ represents the set of known RS scene categories. The target domain is denoted as $\mathcal{D}_t = \{x_j^t\}_{j=1}^{N_t}$, whose samples may belong to either $\mathcal{C}$ or unknown categories. At inference time, all unknown-category samples are assigned to an additional $(C+1)$th label. The objective of OSDA is to learn a discriminative model that recognizes known scenes and rejects unknown scenes using labeled source-domain data and unlabeled target-domain data.

This work adopts a VLM as the foundational framework for semantic modeling of RS scenes. The model consists of an image encoder and a text encoder, which map RS images and category-level textual descriptions into a shared embedding space. Given an RS image x_i, the image encoder extracts a visual feature $f(x_i)$, while the text encoder produces a textual feature $g(k)$ for each known category $k \in \mathcal{C}$. Both visual and textual features are ℓ_2-normalized, and their semantic relevance is measured by cosine similarity

$$\cos(x_i, k) = \frac{f(x_i)^\top g(k)}{\|f(x_i)\|_2 \, \|g(k)\|_2} \tag{1}$$

where $f(x_i)$ and $g(k)$ are the ℓ_2-normalized image and text features, $k \in \{1, 2, \ldots, C\}$. Temperature scaling is used to calibrate the similarity scores, and a softmax operation yields the response probability of the image over the known categories, which is given by

$$p_i(k) = \frac{\exp(\cos(x_i, k)/\tau)}{\sum_{a \in \mathcal{C}} \exp(\cos(x_i, a)/\tau)} \tag{2}$$

where τ denotes the scaling factor. Under the closed-set setting, this distribution can be directly used for classification. In open-set scenarios, however, a single similarity-based distribution is insufficient to characterize samples that do not belong to any known category. Motivated by this limitation, the matching mechanism is extended in the subsequent methodology to better support open-set discrimination for RS scenes.

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B. Positive Multinegative Prompt Module

To explicitly establish category-level binary semantic discrimination in open-set cross-scene classification, we propose a positive multinegative prompt module within a vision–language framework. Different from general computer vision (CV) scenarios, RS scene images usually contain complex elements under significant cross-scene variations, causing unknown categories to exhibit diverse partial semantic similarities to known classes. Therefore, a single negative prompt is insufficient to characterize the diverse nonbelonging semantics in open-set cross-scene RS classification. To address this issue, the proposed framework introduces one group of positive prompts and multiple groups of negative prompts for each known category, so as to jointly model the complementary semantics of belonging-to-class and not-belonging-to-class.

In the text encoder $\mathrm{Enc}_{\mathcal{T}}(\cdot)$, which consists of K transformer blocks, the mapping of the ith block is denoted by $\mathcal{T}^{(i)}(\cdot)$, where $i = 1, \ldots, K$. For each known category $k \in \mathcal{C}$, let $\mathbf{W}_k^{(0)} \in \mathbb{R}^{N \times d_l}$ be the initial token embedding sequence of class k, where N is the token length and d_l is the embedding dimension of the text encoder. At the ith layer, we introduce a set of learnable textual prompt tokens, denoted by $\mathbf{P}^{(i)} = [\mathbf{p}_1^{(i)}, \mathbf{p}_2^{(i)}, \ldots, \mathbf{p}_n^{(i)}] \in \mathbb{R}^{n \times d_l}$, where $\mathbf{p}_j^{(i)} \in \mathbb{R}^{d_l}$ denotes the jth prompt token at the ith layer, and n is the prompt length. The prompt tokens are injected into the first J transformer layers and updated in a layerwise manner as

$$\left[\mathbf{P}^{(i)}, W_k^{(i)}\right] = \mathcal{T}^{(i)}\left(\left[\mathbf{P}^{(i-1)}, W_k^{(i-1)}\right]\right), \quad i = 1, 2, \ldots, J. \tag{3}$$

The remaining layers subsequently transform the output of the preceding layer until the final textual feature $\mathbf{t}$ is obtained, which is derived by extracting the embedding at the end-of-text position and applying a linear projection Proj_t

$$\left[\mathbf{P}^{(j)}, W_k^{(j)}\right] = \mathcal{T}^{(j)}\left(\left[\mathbf{P}^{(j-1)}, W_k^{(j-1)}\right]\right), \quad j = J+1, \ldots, K \tag{4}$$

$$\mathbf{t} = \mathrm{Proj}_t\left(W_k^{(K)}\right). \tag{5}$$

To maintain semantic consistency between the textual and visual prompt spaces, a lightweight cross-modal projection bridge is employed to transform textual prompts into visual prompts

$$\widetilde{\mathbf{P}}^{(i-1)} = \phi^{(i-1)}\left(\mathbf{P}^{(i-1)}\right) \tag{6}$$

where $\phi^{(i-1)}$ denotes a linear projection, and $\widetilde{\mathbf{P}}^{(i-1)}$ denotes the projected visual prompt tokens injected into the image encoder.

For the image encoder $\mathrm{Enc}_{\mathcal{V}}(\cdot)$, let $\mathcal{V}^{(i)}(\cdot)$ denote the mapping of its ith transformer block. The projected visual prompts are injected into the first J transformer blocks and are jointly updated with the class token and patch tokens. The layerwise update is formulated as

$$\left[\mathbf{c}^{(i)}, \mathbf{E}^{(i)}\right] = \mathcal{V}^{(i)}\left(\left[\mathbf{c}^{(i-1)}, \mathbf{E}^{(i-1)}, \widetilde{\mathbf{P}}^{(i-1)}\right]\right) \tag{7}$$

where $\mathbf{c}^{(i)}$ denotes the class token at the ith layer, and $\mathbf{E}^{(i)}$ denotes the corresponding sequence of patch-level features. The remaining transformer blocks further transform only the class token and patch tokens

$$\left[\mathbf{c}^{(j)}, \mathbf{E}^{(j)}, \widetilde{\mathbf{P}}^{(j)}\right] = \mathcal{V}^{(j)}\left(\left[\mathbf{c}^{(j-1)}, \mathbf{E}^{(j-1)}, , \widetilde{\mathbf{P}}^{(j-1)}\right]\right). \tag{8}$$

The final visual feature $\mathbf{v}$ is obtained by applying a linear projection $\mathrm{Proj}_{\mathrm{v}}$ to the class token of the last layer

$$\mathbf{v} = \mathrm{Proj}_{\mathrm{v}}\left(\mathbf{c}^{(K)}\right). \tag{9}$$

Under the multimodal prompt injection framework, we construct one group of positive prompts and M groups of negative prompts at the category level. Specifically, let $\{\mathbf{P}_+^{(i)}\}_{i=0}^{J-1}$ be the learnable textual prompts for the positive branch, and let $\{\mathbf{P}_{m,-}^{(i)}\}_{i=0}^{J-1}$ be the learnable textual prompts for the mth negative branch, where $m = 1, \ldots, M$. For category $k \in \mathcal{C}$, the corresponding positive and negative textual features are defined as

$$\mathbf{t}_k^+ = \mathrm{Enc}_{\mathcal{T}}\left(\mathbf{W}_k^{(0)}; \mathbf{P}_+^{(i)}\right), \quad \mathbf{t}_{k,m}^- = \mathrm{Enc}_{\mathcal{T}}\left(\mathbf{W}_k^{(0)}; \mathbf{P}_{m,-}^{(i)}\right). \tag{10}$$

According to (6), the projected visual prompts corresponding to the positive branch and the mth negative branch are denoted by $\{\widetilde{\mathbf{P}}_+^{(i)}\}_{i=0}^{J-1}$ and $\{\widetilde{\mathbf{P}}_{m,-}^{(i)}\}_{i=0}^{J-1}$, respectively. For an input RS image x, the resulting branch-specific visual features are defined as

$$\mathbf{v}^+ = \mathrm{Enc}_{\mathcal{V}}\left(x; \widetilde{\mathbf{P}}_+^{(i)}\right), \quad \mathbf{v}_m^- = \mathrm{Enc}_{\mathcal{V}}\left(x; \widetilde{\mathbf{P}}_{m,-}^{(i)}\right). \tag{11}$$

With both encoders kept frozen, the matching scores of the positive branch and the mth negative branch with respect to category k are computed as

$$s_k^+(x) = \cos\left(\mathbf{v}^+, \mathbf{t}_k^+\right), \quad s_{k,m}^-(x) = \cos\left(\mathbf{v}_m^-, \mathbf{t}_{k,m}^-\right). \tag{12}$$

As a result, a positive score vector $s^+(x) \in \mathbb{R}^C$ and a multigroup negative score tensor $s^-(x) \in \mathbb{R}^{C \times M}$ are obtained. During training, to align with the binary structure of the open-set loss, the multiple negative scores are aggregated along the prompt dimension as $\bar{s}_k(x) = 1/M \sum_{m=1}^{M} s_{k,m}^-(x)$. Multiple groups of negative prompts characterize discriminative semantic cues along the direction of not belonging to a class from diverse semantic perspectives, such that nonbelonging semantics are no longer restricted to a single semantic exclusion pattern. Through aggregation along the prompt dimension, the model can obtain a more robust and semantically comprehensive negative reference, which effectively alleviates the susceptibility of a single negative prompt to semantic bias and noise under cross-domain and open-set scenarios. The positive prompts mainly encode category-level belonging semantics, whereas the multiple negative prompts encode diverse nonbelonging semantic patterns associated with category exclusion under open-set cross-scene RS conditions. Therefore, the learned prompts can be understood as task-adaptive semantic directions that guide the model to establish discriminative positive and negative semantic responses for known-category recognition and unknown-sample rejection. The resulting aggregated negative semantic score $\bar{s}_k(x)$ serves as a stable openness-aware reference term and, together with $s_k^+(x)$, jointly supports the unified optimization of closed-set supervision and open-set supervision on the source domain, as well as open-set regularization on the target domain.

C. Positive–Negative Semantic-Guided Known–Unknown Separation

This section presents the proposed PNKUS strategy, as illustrated in Fig. 4. In open-set cross-scene RS domain adaptation,

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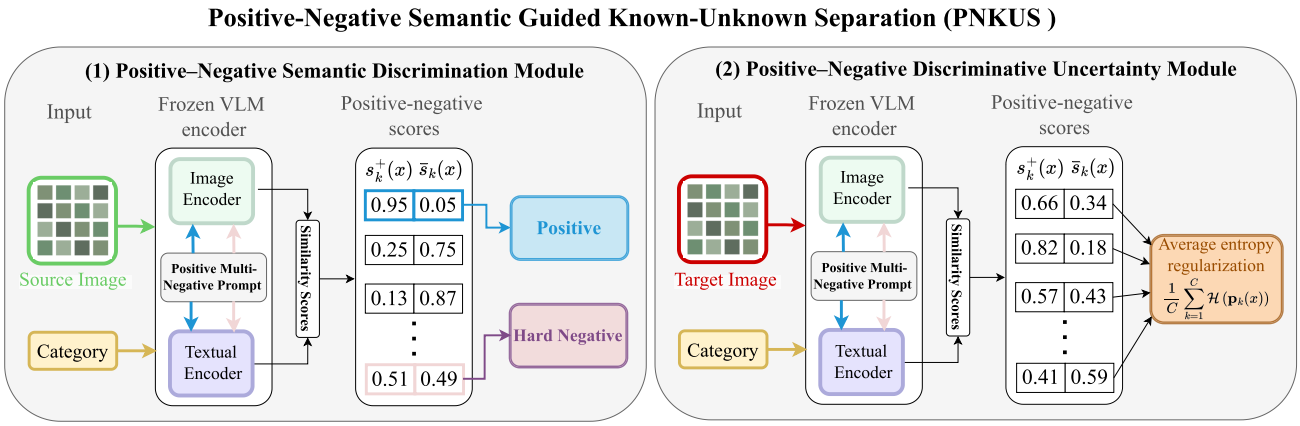


Fig. 4. Proposed PNKUS strategy consists of a source-domain positive–negative semantic discrimination module and a target-domain discriminative uncertainty module. The source-domain module establishes category-level binary semantic boundaries by jointly modeling positive prompts and multiple negative prompts and further strengthens known-class discrimination through hard negative mining. The target-domain module converts category-level positive–negative semantic responses into binary probability distributions and optimizes them with average entropy regularization, thereby reducing the disturbance of unknown samples and promoting robust known–unknown separation.

the target domain contains both shared known categories and unknown categories under significant cross-scene distribution shifts, making it difficult to achieve reliable known–unknown separation using only positive semantic matching. To address this issue, PNKUS constructs, for each category $k \in \mathcal{C}$, a binary semantic response pair $[s_k^+(x), \bar{s}_k(x)]$, where $s_k^+(x)$ represents the positive semantic responses that sample x belongs to category k, and $\bar{s}_k(x)$ denotes the aggregated negative semantic responses that it does not belong to category k. Based on this category-level positive–negative semantic structure, PNKUS consists of a source-domain positive–negative semantic discrimination module and a target-domain discriminative uncertainty module.

1) Positive–Negative Semantic Discrimination Module: In the source domain, the proposed strategy combines closed-set classification supervision with category-level positive–negative semantic discrimination. Based on the positive prompt branch, the closed-set classification probability over the known category set is defined as $p_k^+(x) = \exp(s_k^+(x))/\sum_{j=1}^{C}\exp(s_j^+(x))$. Given a labeled source-domain sample (x^s, y^s) with $y^s \in \mathcal{C}$, the closed-set classification loss is formulated as

$$\mathcal{L}_{cls} = -\log p_{y^s}^+(x). \tag{13}$$

This term provides standard closed-set supervision for the positive branch and encourages discriminative cross-modal representations for shared known categories. Beyond closed-set supervision, we further impose a positive–negative semantic discriminative constraint to explicitly establish category-level binary semantic boundaries of belonging and nonbelonging. Specifically, for each category k, the model produces a positive matching score $s_k^+(x)$ and an aggregated negative semantic score $\bar{s}_k(x)$. For the ground-truth category y^s, the desired relation is $s_{y^s}^+(x) > \bar{s}_{y^s}(x)$, such that the positive semantic responses dominate the corresponding negative responses. For non-ground-truth categories, the desired relation is reversed, namely, $\bar{s}_k(x) > s_k^+(x)$, indicating nonbelonging semantics.

Considering that different non-ground-truth categories may exhibit different levels of semantic confusion with the ground-truth category, we employ hard negative mining and only optimize the most confusing negative category. The hardest negative category is defined as $k^* = \arg\max_{k\neq y^s}(s_k^+(x) - \bar{s}_k(x))$, which corresponds to the non-ground-truth category whose positive responses are most competitive relative to its negative responses. Based on this, the source-domain open-set discriminative loss is formulated as

$$\mathcal{L}_{\text{open}} = -\log\frac{\exp\left(s_{y^s}^+(x)\right)}{\exp\left(s_{y^s}^+(x)\right)+\exp\left(\bar{s}_{y^s}(x)\right)} - \log\frac{\exp\left(\bar{s}_{k^*}(x)\right)}{\exp\left(s_{k^*}^+(x)\right)+\exp\left(\bar{s}_{k^*}(x)\right)}. \tag{14}$$

The first term enforces positive-semantic dominance for the ground-truth category, while the second term enforces negative-semantic dominance for the hardest non-ground-truth category. In this way, the model directly learns a category-level binary semantic boundary in the source domain. Meanwhile, the use of multiple groups of negative prompts alleviates single-semantic bias and provides complementary nonbelonging cues. As a result, the source-domain module learns category-level nonbelonging semantics and establishes an open-set discrimination prior for subsequent known–unknown separation in the target domain.

2) Positive–Negative Discriminative Uncertainty Module: To further support robust OSDA under unlabeled target conditions, we formulate a target-domain objective in the category-level positive–negative binary semantic space. For a target-domain sample x^t, the model produces, for each known category $k \in \{1, \ldots, C\}$, a pair of positive and negative semantic responses $[s_k^+(x), \bar{s}_k(x)]$, which, respectively, indicate the semantic responses of belonging to category k and not belonging to category k. Based on these two responses, a category-level binary probability distribution is obtained as $\mathbf{p}_k(x) = \text{softmax}([s_k^+(x), \bar{s}_k(x)])$. Here, the two components of $\mathbf{p}_k(x)$ represent the probabilities of belonging-to-class and not-belonging-to-class, respectively, thereby characterizing

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the binary semantic tendency of sample x with respect to category k.

Under the unsupervised setting, if a target sample belongs to a known category, it is expected to exhibit relatively confident positive–negative semantic responses for the corresponding category. In contrast, potential unknown samples, due to the lack of explicit positive semantic support, tend to present higher uncertainty across multiple categories. Motivated by this observation, the proposed approach suppresses the diffusion of predictive uncertainty by minimizing the entropy [40] of the above binary semantic distribution. This strategy guides the model to form a clearer open-set discriminative structure in the target domain. The uncertainty regularization term for the target domain is defined as

$$\mathcal{L}_{ent} = \frac{1}{C}\sum_{k=1}^{C}\mathcal{H}\left(\mathbf{p}_k\left(x\right)\right) \tag{15}$$

where $\mathcal{H}(\cdot)$ denotes the entropy function. It is worth emphasizing that this uncertainty regularization does not rely on any pseudo-labels or threshold settings. Instead, it is directly imposed on the positive–negative semantic contrast space and constrains the discriminative behavior of target-domain samples at the category level. Through joint optimization with the positive–negative discrimination structure established in the source domain, this module suppresses semantic interference introduced by unknown samples during cross-domain alignment while preserving discriminative consistency for known categories.

D. Semantic Regularization of Positive–Negative Prompts

1) Semantic Diversity Regularization for Negative Prompts: In the positive multinegative prompt module, multiple groups of negative prompts are intended to characterize the concept of not belonging to a class from different semantic perspectives. However, without explicit constraints, different negative prompts may converge to highly similar representations during optimization, which weakens the diversity and coverage of negative semantics. To prevent semantic collapse among multiple negative prompts within the same category, this work encourages sufficient semantic disparity among category-specific negative prompts in the semantic space. Specifically, let $\mathbf{t}_{k,m}^{-}$ be the textual feature corresponding to the mth negative prompt of category k. To prevent multiple negative prompts of the same category from collapsing to nearly identical semantics, we impose a pairwise similarity regularization within each category. The cosine similarity between two negative prompts is computed as $\mathrm{sim}_k(m,m') = (\mathbf{t}_{k,m}^{-})^{\top}\mathbf{t}_{k,m'}^{-}/\|\mathbf{t}_{k,m}^{-}\|_2\|\mathbf{t}_{k,m'}^{-}\|_2, \quad m \neq m'$. Based on this similarity, we introduce a diversity regularization among negative prompts within each category as

$$\mathcal{L}_{nn} = \frac{1}{C\cdot M\cdot(M-1)}\sum_{k=1}^{C}\sum_{m\neq m'}\left[\max\left(0,\mathrm{sim}_k\left(m,m'\right)-\delta\right)\right]^2. \tag{16}$$

This term penalizes only those negative-prompt pairs whose similarity exceeds the tolerance threshold δ. In other words, moderate similarity is allowed, since multiple negative prompts of the same category should still share category-level nonbelonging semantics. The penalty is activated only when two negative prompts become overly similar, indicating semantic redundancy. The squared form further increases the penalty for highly redundant pairs and stabilizes optimization. As a result, $\mathcal{L}_{nn}$ encourages multiple negative prompts within the same category to encode complementary rather than duplicated nonbelonging semantics. This improves the diversity and robustness of negative semantic modeling, thereby providing a more expressive basis for unknown-class characterization.

2) Discriminative Margin Regularization Between Positive and Negative Prompts: Beyond promoting diversity among negative prompts, it is also necessary to maintain a reasonable semantic margin between negative prompts and the corresponding positive prompt of each category. If the semantics of negative prompts become excessively close to those of the positive prompt, positive and negative semantic scores become difficult to distinguish, which in turn degrades category-level discriminative capability. To address this issue, an upper bound constraint is imposed on the semantic similarity between positive and negative prompts within the same category. Let $\mathbf{t}_k^{+}$ be the textual feature of the positive prompt for category k, and let $\mathbf{t}_{k,m}^{-}$ be the textual feature of the mth negative prompt. Specifically, the cosine similarity between positive and negative prompts is computed as $\mathrm{sim}_{k,m}^{pn} = (\mathbf{t}_k^{+})^{\top}\mathbf{t}_{k,m}^{-}/\|\mathbf{t}_k^{+}\|_2\|\mathbf{t}_{k,m}^{-}\|_2$. A penalty term is imposed only when the similarity exceeds the threshold δ, and the corresponding loss is defined as

$$\mathcal{L}_{pn} = \frac{1}{C\cdot M}\sum_{k=1}^{C}\sum_{m=1}^{M}\max\left(0,\mathrm{sim}_{k,m}^{pn}-\delta\right). \tag{17}$$

This constraint encourages the semantics of negative prompts to remain distributed outside a reasonable neighborhood of the positive prompt semantics, thereby maintaining a clear positive–negative discriminative margin in the semantic space and providing stable support for open-set rejection.

Based on the above components, the overall loss of the proposed method is formulated as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{cls} + \mathcal{L}_{open} + \mathcal{L}_{ent} + \alpha\mathcal{L}_{nn} + \beta\mathcal{L}_{pn}. \tag{18}$$

Here, α and β are the weighting factors that balance the negative prompt diversity regularization loss and the positive–negative discriminative margin regularization loss, respectively.

E. Open-Set Inference

During the testing stage, the model is required to simultaneously perform known RS scene classification and unknown sample rejection on target-domain samples. Unlike open-set discrimination strategies that rely on manually specified thresholds, this work designs a threshold-free open-set inference mechanism based on the relative semantic discrepancy between positive and negative semantic scores. For a given test sample x, the model outputs, for each known category $k \in \{1,\ldots,C\}$, a positive matching score $s_k^{+}(x)$ and the corresponding negative score $\bar{s}_k(x)$. These two scores, respectively, reflect the supportive and exclusive semantic relationships

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between the sample and the category. Based on this formulation, a category-level binary semantic difference is defined as $\Delta_k(x) = s_k^+(x) - \bar{s}_k(x)$. It measures the binary semantic margin between belonging and nonbelonging with respect to category k. When $\Delta_k(x)$ is large, the sample is more likely to belong to the corresponding category. Conversely, a smaller value indicates that the negative response dominates. Among all known categories, the category with the maximum semantic dominance is first selected as

$$k^* = \arg\max_{k\in\{1,\ldots,C\}} \Delta_k(x). \tag{19}$$

Subsequently, the decision is made according to the sign of the maximum semantic difference. If $\Delta_k(x) > 0$, the sample is predicted as the known category k^*. Otherwise, the sample is considered to lack sufficient positive semantic support across all known categories and is therefore uniformly assigned to the unknown class. Formally, the final prediction is defined as

$$\hat{y}(x) = \begin{cases} k^*, & \text{if } \max\ \Delta_k(x) > 0 \\ C+1, & \text{otherwise}. \end{cases} \tag{20}$$

Here, the (C + 1)th class represents a unified unknown label that is used to reject all unknown samples rather than to perform fine-grained recognition among unknown categories. This threshold-free inference mechanism makes decisions solely based on the relative relationship between positive and negative semantic responses, without relying on additional threshold searching.

The overall training procedure of the proposed CLPNP framework is summarized in Algorithm 1.

Algorithm 1 Training Procedure of CLPNP

Input: Labeled source domain $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ (known categories $\mathcal{C}$); Unlabeled target domain $\mathcal{D}_t = \{x_j^t\}_{j=1}^{N_t}$; Hyperparameters: α, β, prompt length n, prompt depth J, negative prompt groups M
Output: Learnable positive prompt $\mathbf{P}_+$ and M groups of negative prompt $\{\mathbf{P}_{m,-}\}_{m=1}^{M}$
Set hyperparameters: $\alpha = 1.0$, $\beta = 1.0$, $n = 2$, $J = 9$, $M = 4$
for *epoch* = 1 **to** *num_epoch* **do**
    Sample source batch $\mathcal{B}_s$ and target batch $\mathcal{B}_t$ from $\mathcal{D}_s$ and $\mathcal{D}_t$, respectively.
    Calculate positive prompt-based features $(\mathbf{v}^+, \mathbf{t}_k^+)$ and negative prompt-based features $(\mathbf{v}_m^-, \mathbf{t}_{k,m}^-)$ using Eq. (10) and Eq. (11).
    Compute positive scores $s_k^+(x)$ and negative scores $s_{k,m}^-(x)$ following Eq. (12), then aggregate negative scores as $\bar{s}_k(x) = \frac{1}{M}\sum_{m=1}^{M} s_{k,m}^-(x)$.
    Compute source domain semantic discrimination losses: closed-set classification loss $\mathcal{L}_{\text{cls}}$ (Eq. (13)) and open-set discrimination loss $\mathcal{L}_{\text{open}}$ (Eq. (14)).
    Compute target domain uncertainty regularization loss $\mathcal{L}_{\text{ent}}$ (Eq. (15)).
    Compute semantic regularization losses: negative prompt diversity loss $\mathcal{L}_{\text{nn}}$ (Eq. (16)) and positive–negative margin loss $\mathcal{L}_{\text{pn}}$ (Eq. (17)).
    Calculate total loss $\mathcal{L}_{\text{total}}$ (Eq. (18)), then update $\mathbf{P}_+$ and $\mathbf{P}_{m,-}$ via backpropagation to minimize $\mathcal{L}_{\text{total}}$.
end for

IV. Experiments

This section presents the experimental settings and datasets, followed by quantitative and qualitative results and an in-depth analysis of the findings.

A. Datasets

To comprehensively and rigorously evaluate the effectiveness of the proposed method, experiments are conducted on six publicly available RS scene classification benchmark datasets. These datasets exhibit substantial diversity in data scale, spatial resolution, and sample distribution, thereby providing realistic and challenging test conditions for a thorough assessment of the model's adaptation capability and classification accuracy.

1) *AID (A) [41]:* This dataset is a large-scale benchmark for aerial scene classification, released in 2017 by Wuhan University and several collaborating institutions. It contains a total of 10 000 RGB aerial images and occupies approximately 2.6 GB. All images have a spatial resolution of 600 × 600 pixels, and the dataset comprises 30 scene categories, with 200–400 images in each category.
2) *UCM (U) [42]:* All images in the UC Merced dataset are extracted from U.S. Geological Survey National Map Urban Area Imagery collection. The dataset has an overall size of approximately 317.07 MB and consists of 21 land-use categories, with 100 images per category, resulting in a total of 2100 images. Each image has a spatial resolution of 256 × 256 pixels.
3) *NWPU (N) [43]:* This dataset was released by Cheng et al. [43] from Northwestern Polytechnical University. It contains 31 500 RGB RS images with a spatial resolution of 256 × 256 pixels. The dataset encompasses 45 scene categories, each comprising 700 images.
4) *CLRS (C) [44]:* This dataset contains 15 000 RS images across 25 scene categories, with 600 images per category. All images are of size 256 × 256 pixels. The spatial resolution spans a wide range from 0.26 to 8.85 m, allowing the dataset to realistically simulate RS observations under varying imaging conditions.
5) *MLRSN (M) [45]:* The dataset is primarily based on Sentinel-2 multispectral imagery. It comprises 109 161 high-resolution RS images annotated across 46 scene categories, with each category containing between 1500 and 3000 images. All images are standardized to a size of 256 × 256 pixels, encompassing a variety of spatial resolutions.
6) *OPTIMAL-31 (O) [46]:* Released by Northwestern Polytechnical University in 2019, this dataset consists of 1860 images spanning 31 scene categories, with 60 images per category. All images are 256 × 256 pixels in size and are stored in three-channel JPEG format.

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TABLE I
CLASS DISTRIBUTION IN THE AID_UCM_NWPU SETTING

No.	Class	AID	UCM	NWPU
1	Airfield	360	100	1400
2	Anchorage	380	100	700
3	Beach	400	100	700
4	Dense_Residential	410	100	700
5	Farm	370	100	1400
6	Flyover	420	100	700
7	Forest	250	100	700
8	Game_Space	660	100	1400
9	Parking_Space	390	100	700
10	River	410	100	700
11	Sparse_Residential	300	100	700
12	Storage_Cisterns	360	100	700
	Count	4710	1200	10500

TABLE II
CLASS DISTRIBUTION IN THE AID_CLRS_MLRSN_OPT SETTING

No.	Class	AID	CLRS	MLRSN	OPT
1	Airport	180	120	110	60
2	Beach	200	120	125	60
3	Bridge	180	120	125	60
4	Commercial	175	120	125	60
5	Desert	150	120	127	60
6	Farmland	185	120	126	120
7	Forest	125	120	126	60
8	Industrial	195	120	106	60
9	Meadow	140	120	126	60
10	Mountain	170	120	125	60
11	Overpass	210	120	125	60
12	Parking	195	120	126	60
13	Playground	185	120	125	60
14	Port	190	120	126	60
15	Residential	205	120	140	60
	Count	2685	1800	1863	960

B. Experimental Setup

To thoroughly evaluate the proposed method, two groups of cross-scene RS classification experimental settings are constructed, resulting in a total of 18 cross-scene classification tasks evaluated using four metrics.

First, the AID, UC Merced, and NWPU datasets are selected to construct the experimental setting. From these three datasets, 12 scene categories shared across datasets are identified, as summarized in Table I. An open-set protocol is adopted in which the first nine categories are treated as shared known categories, while the remaining three categories are regarded as target private unknown categories. Each dataset is considered as an independent domain, and source–target domain pairs are constructed accordingly. This setting results in six unidirectional cross-domain tasks, namely, A→U, A→N, U→A, U→N, N→A, and N→U.

Second, the AID, CLRS, MLRSN, and OPTIMAL-31 datasets are selected to construct another experimental setting. The 15 scene categories common to all four datasets are extracted as the base label set, as listed in Table II. These categories are then randomly partitioned into seven known categories and eight target-private unknown categories, where the latter are treated as unknown and highlighted in blue. Each dataset serves as an independent domain, with 12 cross-domain transfer tasks formed by pairwise combinations: A→C, A→M, A→O, C→A, C→M, C→O, M→A, M→C, M→O, O→A, O→C, and O→M.

To comprehensively evaluate model performance in the OSDA setting, four evaluation metrics are adopted, namely, OS*, UNK, HOS, and ALL [31]. Specifically, OS* denotes the mean per-class accuracy over the shared known categories in the target domain and is used to assess the recognition capability on known categories. UNK refers to unknown accuracy, which measures the accuracy of identifying unknown samples in the target domain and reflects the model's ability to distinguish unknown categories. Since OSDA requires a proper balance between known-class recognition and unknown-class detection, the harmonic mean of OS* and UNK, denoted as HOS, is further employed as a comprehensive metric. It is defined as

$$HOS = \frac{2 \times OS^* \times UNK}{OS^* + UNK} \tag{21}$$

which jointly characterizes the model's overall performance on both known and unknown categories. In addition, ALL denotes the overall accuracy on the target-domain test set, where a prediction is considered correct if a known sample is correctly classified or an unknown sample is correctly rejected. Furthermore, in Section IV-D, we adopt the false positive rate (FPR) as an evaluation metric, which is defined as the proportion of unknown class samples incorrectly classified as known-class samples.

All experiments are conducted on a workstation equipped with an NVIDIA GeForce RTX 4090 GPU and implemented using the PyTorch 2.0.0 framework. The CLIP ViT-B/16 model is adopted as the backbone network. The textual prompt context is initialized as "This is a satellite image of," with a prompt length of two tokens and a prompt depth of 9. To enhance the discriminative capability for unknown categories, multiple groups of negative prompts are constructed for each known category, with the number of negative prompt groups set to M = 4, and the positive and negative prompts are jointly leveraged during training. During optimization, the loss weights are set to $\alpha = 1.0$ and $\beta = 1.0$. All input images are resized to 224×224 pixels prior to training. The model is trained for 35 epochs using the SGD optimizer, with an initial learning rate of 0.0035 and a cosine learning rate scheduling strategy.

C. Experimental Results

To comprehensively evaluate the effectiveness of the proposed approach in OSDA settings for RS, a systematic comparison is conducted against a range of representative OSDA methods. In total, ten existing methods are included in the comparison, namely, OSBP [11], DAMC [14], UADAL [12], OVANet [47], SSOUDA [15], MAOSDAN [16], OKRA [34], PUMCL [17], OSClip [21], and TASC [48]. Among them, methods specifically developed for RS scene

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TABLE III
COMPARISON WITH OSDA METHODS IN THE AID_UCM_NWPU SETTING

Task	Metric	Method										
		OSBP	DAMC	UADAL	OVANet	SSOUDA	MAOSDAN	OKRA	PUMCL	OSClip	TASC	OURS
A→U	OS*	86.56	76.33	55.56	76.44	81.78	75.67	73.11	**88.22**	82.11	73.44	**86.78**
	UNK	45.33	61.00	78.65	76.00	47.00	**97.66**	91.00	91.33	95.99	**100.00**	**97.66**
	HOS	59.50	67.81	65.12	76.22	59.69	85.27	81.08	**89.75**	88.51	84.69	**91.90**
	ALL	76.25	72.50	61.33	76.33	73.08	81.17	77.58	**88.53**	83.50	80.08	**89.50**
A→N	OS*	79.24	68.83	76.44	82.05	**90.98**	71.48	**92.50**	74.81	84.09	73.93	89.21
	UNK	64.76	66.47	88.57	79.86	76.48	**96.70**	92.62	89.48	86.71	**99.20**	88.75
	HOS	71.27	67.63	82.06	80.94	83.10	82.20	**92.56**	81.49	85.38	84.72	**88.98**
	ALL	73.10	64.47	74.89	79.37	86.87	72.24	**90.80**	76.27	84.35	76.77	**89.19**
U→A	OS*	82.52	57.96	50.67	48.97	62.33	66.69	**83.66**	73.64	80.96	74.91	**85.04**
	UNK	58.79	73.19	80.19	81.50	63.54	**97.28**	94.77	83.64	95.33	79.72	**95.53**
	HOS	68.66	64.69	62.10	61.18	62.93	79.13	**88.87**	78.32	87.56	77.24	**89.98**
	ALL	77.35	60.42	55.12	55.31	62.31	71.63	**86.58**	74.31	82.40	70.40	**85.05**
U→N	OS*	73.57	58.11	60.08	54.06	70.61	59.27	**84.34**	70.55	82.09	73.70	**87.76**
	UNK	43.55	73.62	81.78	91.95	69.77	**98.05**	91.86	89.52	93.47	**97.86**	97.62
	HOS	54.71	64.95	69.27	68.09	70.19	73.88	**87.94**	78.91	87.41	84.08	**92.43**
	ALL	66.34	56.22	56.13	55.02	65.47	64.11	80.88	72.45	**83.23**	75.89	**85.42**
N→A	OS*	87.08	72.76	85.26	92.54	92.74	77.18	**93.64**	85.54	89.89	76.08	**96.31**
	UNK	56.26	79.15	95.14	63.46	61.96	**99.63**	77.38	97.02	87.02	**99.80**	89.91
	HOS	68.36	75.82	89.93	75.29	74.29	86.98	84.74	**90.92**	88.43	86.34	**93.00**
	ALL	78.62	73.18	85.92	84.97	84.59	81.46	88.47	86.69	**89.93**	76.62	**95.07**
N→U	OS*	65.00	73.18	70.89	**93.44**	**96.33**	77.33	85.56	80.11	83.06	78.11	89.00
	UNK	36.33	68.44	83.33	69.00	45.00	**99.33**	78.34	91.32	90.48	91.34	**100.00**
	HOS	46.61	70.73	76.61	79.38	61.34	**86.96**	81.79	85.35	86.61	84.21	**94.18**
	ALL	57.83	71.67	74.00	87.33	72.25	82.83	83.75	81.23	**88.78**	81.42	**91.75**
Avg.	OS*	79.00	67.86	66.48	74.58	82.46	71.27	**85.47**	78.81	83.70	75.03	**89.02**
	UNK	50.84	70.31	84.61	76.96	60.62	**98.11**	87.66	90.39	91.50	94.65	**94.91**
	HOS	61.52	68.61	74.18	73.52	68.59	82.40	86.16	84.12	**87.32**	83.55	**91.75**
	ALL	71.58	66.41	67.90	73.06	74.10	75.57	84.68	79.91	**85.37**	76.86	**89.33**

classification include SSOUDA [15], MAOSDAN [16], OKRA [34], PUMCL [17], and OSClip [21]. Approaches based on the ViT-B backbone include OSClip [21] and TASC [48], while the remaining methods adopt ResNet-50 as the visual backbone. It should be noted that OSDA methods built upon VLM backbones such as ViT-B remain relatively limited. As a result, only OSClip [21] and TASC [48] are selected as representative baselines under the VLM setting to ensure a fair and meaningful comparison.

Table III presents the OS*, UNK, HOS, and ALL results in the AID_UCM_NWPU setting. The proposed method achieves the highest OS* (89.02%) and ALL (89.33%), outperforming OKRA [34] (85.47%) and OSClip [21] (85.37%) by 3.55% and 3.96%, respectively, indicating improved known-category recognition and overall performance. Compared with these methods, OKRA [34] relies on discriminative responses and uncertainty estimation, making its unknown rejection sensitive to score distributions and feature drift. OSClip [21] enhances cross-scene generalization via positive semantic matching, but its unknown discrimination still depends on thresholding or postprocessing. On HOS, the proposed method achieves 91.75%, exceeding OSClip [21] (87.32%) by 4.43%, reflecting a more favorable balance between known-category recognition and unknown-sample rejection. Although UNK is not the highest, methods such as MAOSDAN [16] obtain higher UNK through aggressive rejection strategies that sacrifice known-class accuracy, resulting in lower HOS. Overall, these results indicate that the proposed PNKUS strategy achieves a more balanced open-set performance under cross-scene conditions.

Table IV further evaluates the proposed method under the more challenging AID_CLRS_MLRSN_OPT setting, which involves larger variations in scene scale and category distributions. The proposed method achieves strong average performance across all metrics, obtaining 88.50% on OS*, 90.17% on UNK, 89.21% on HOS, and 89.47% on ALL. Notably, it consistently outperforms existing VLM-based approaches, improving OS* over OSClip [21] by 0.54%, UNK over TASC [48] by 1.57%, HOS over OSClip by 1.82%, and ALL by 1.59%. Compared with representative VLM-based methods such as OSClip and TASC, which mainly rely on single positive prompts to encode class-belonging semantics, the proposed approach explicitly introduces category-level negative prompts to model class-nonbelonging semantics. Without explicit semantic descriptions for unknown categories, single positive prompting is inherently insufficient to characterize the semantic diversity of unknown samples under complex cross-scene shifts. By contrast, the proposed method captures richer open-set semantics through positive–negative prompt contrast, enabling more reliable known–unknown separation across datasets, as consistently reflected by its superior HOS performance.

D. Experimental Analysis

In this section, seven ablation studies are presented to validate the effectiveness of the proposed method.

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TABLE IV
COMPARISON WITH OSDA METHODS IN THE AID_CLRS_MLRSN_OPT SETTING

Task	Metric	Method										
		OSBP	DAMC	UADAL	OVANet	SSOUDA	MAOSDAN	OKRA	PUMCL	OSClip	TASC	OURS
A→C	OS*	66.19	70.95	70.12	81.79	**87.50**	71.79	**90.12**	78.21	80.71	75.00	82.50
	UNK	78.83	69.66	**89.00**	57.61	39.16	87.67	62.40	85.52	**90.22**	83.34	83.75
	HOS	71.96	70.30	78.44	67.60	54.11	78.94	73.74	81.70	**85.20**	78.95	**83.12**
	ALL	67.89	69.61	74.11	68.89	70.22	75.06	75.33	79.12	**81.90**	79.44	**83.17**
A→M	OS*	55.58	55.61	63.70	75.18	64.24	63.73	**84.22**	64.10	80.21	72.31	**85.01**
	UNK	65.87	62.36	66.98	57.57	42.74	77.84	55.28	65.56	81.28	**84.66**	**89.15**
	HOS	60.29	58.79	65.30	65.21	51.33	70.08	66.75	64.82	**80.74**	78.00	**87.03**
	ALL	57.16	56.24	64.48	65.81	57.05	67.49	68.55	64.28	**80.84**	79.66	**87.33**
A→O	OS*	79.23	72.86	78.45	**95.71**	88.93	82.02	92.38	81.19	**97.50**	80.83	94.40
	UNK	69.33	64.99	75.66	49.37	37.00	91.34	60.00	82.66	84.16	**97.92**	**91.87**
	HOS	73.95	68.70	77.03	65.14	52.26	86.43	72.75	81.92	**90.34**	88.56	**93.12**
	ALL	69.81	67.78	71.02	72.50	69.81	79.54	76.46	81.38	**95.83**	90.52	**93.02**
C→A	OS*	80.77	82.52	83.15	90.91	96.84	70.15	**97.75**	78.83	79.13	89.06	**97.26**
	UNK	68.72	74.18	85.59	54.57	43.69	85.26	68.43	**97.56**	**98.58**	88.14	92.29
	HOS	74.26	78.13	84.35	68.20	60.21	76.97	80.50	87.20	87.79	**88.60**	**94.71**
	ALL	74.85	80.29	83.42	71.99	79.24	72.25	82.42	81.17	**89.09**	88.08	**94.60**
C→M	OS*	57.12	55.01	52.20	68.77	71.35	61.65	78.45	62.55	**86.26**	67.77	**81.69**
	UNK	67.15	56.94	70.32	55.08	34.44	80.54	55.68	72.41	**91.54**	90.44	**91.73**
	HOS	61.73	55.96	59.92	61.17	46.46	69.84	65.13	67.12	**88.82**	77.48	**86.42**
	ALL	58.67	56.84	57.97	61.73	55.60	66.36	66.34	63.79	**86.92**	80.30	**87.28**
C→O	OS*	61.61	66.07	58.10	87.26	80.36	71.31	**95.36**	72.62	94.05	75.95	**94.29**
	UNK	69.66	63.00	78.65	52.08	47.00	87.01	67.08	80.00	88.95	**94.99**	**92.08**
	HOS	65.39	64.50	66.83	65.23	59.31	78.38	78.76	76.13	**91.43**	84.41	**93.17**
	ALL	59.72	60.65	58.06	70.10	68.43	67.69	81.35	73.54	**91.46**	86.56	**93.44**
M→A	OS*	64.95	61.06	60.37	70.44	81.85	46.05	81.43	78.62	**96.86**	**95.65**	92.26
	UNK	63.68	50.73	80.10	50.28	44.24	59.56	51.21	68.83	83.28	**89.57**	**90.50**
	HOS	64.31	55.42	68.85	58.68	57.44	51.94	62.88	73.40	89.56	**92.51**	**91.37**
	ALL	63.64	54.70	66.88	59.55	67.86	48.14	65.44	77.39	89.65	**92.40**	**91.25**
M→C	OS*	52.92	45.24	48.93	54.05	68.33	43.33	59.29	54.29	**83.81**	65.24	**80.36**
	UNK	55.67	67.34	55.01	47.71	28.83	72.17	40.00	65.50	80.82	**88.94**	**81.24**
	HOS	54.26	54.12	51.79	50.68	40.55	54.15	47.77	59.37	**82.29**	75.27	**80.80**
	ALL	52.28	51.11	47.56	50.67	50.39	51.44	49.00	55.69	**82.22**	77.89	**80.83**
M→O	OS*	70.54	67.38	66.19	84.05	77.14	64.52	83.81	72.62	**85.12**	81.31	**94.64**
	UNK	69.67	73.66	78.00	48.12	43.67	82.34	52.50	76.33	**95.20**	**88.34**	87.30
	HOS	70.10	70.38	71.61	61.20	55.77	72.35	64.56	74.43	**89.88**	84.68	**90.82**
	ALL	63.89	63.15	62.04	61.88	64.07	63.15	63.23	73.08	**86.88**	85.73	**91.04**
O→A	OS*	75.39	68.24	79.78	89.16	90.94	68.84	96.22	76.29	**97.99**	95.58	**97.82**
	UNK	44.58	73.65	92.52	42.00	51.73	**92.98**	62.64	79.56	85.15	90.71	**97.22**
	HOS	56.03	70.84	85.68	57.10	65.95	79.11	75.88	77.89	91.12	**93.08**	**97.52**
	ALL	63.21	68.54	82.92	64.47	77.84	72.40	78.62	76.70	**96.38**	92.96	**97.43**
O→C	OS*	59.88	60.12	56.90	67.50	66.67	66.31	75.24	62.62	**90.71**	**83.81**	83.57
	UNK	58.01	61.50	**83.01**	58.43	37.83	64.67	60.42	79.33	80.00	75.94	**89.69**
	HOS	58.93	60.80	67.52	62.64	48.27	65.48	67.02	69.99	**85.02**	79.68	**86.52**
	ALL	56.89	59.11	62.17	62.67	56.17	63.44	67.33	64.71	**89.37**	79.61	**86.83**
O→M	OS*	62.31	61.30	61.83	80.41	80.87	64.84	**86.09**	75.96	**83.18**	73.89	78.24
	UNK	44.02	52.16	83.25	53.78	44.02	88.82	48.21	75.48	90.14	**90.24**	**95.22**
	HOS	51.59	56.36	70.96	64.45	57.01	74.96	61.81	75.72	**86.52**	81.25	**85.90**
	ALL	56.57	58.72	68.73	66.02	68.73	73.57	65.54	78.92	**84.05**	81.70	**87.39**
Avg.	OS*	65.54	63.86	64.98	78.77	79.59	64.55	85.03	71.49	**87.96**	79.70	**88.50**
	UNK	62.93	64.18	78.17	52.22	41.20	80.85	56.99	77.39	87.44	**88.60**	**90.17**
	HOS	63.57	63.69	70.69	62.28	54.06	71.55	68.13	74.14	**87.39**	83.54	**89.21**
	ALL	62.05	62.23	66.61	64.69	65.45	66.71	69.97	72.48	**87.88**	84.57	**89.47**

1) Loss Ablation Study: Table V reports the ablation results obtained by removing individual loss components in the AID_UCM_NWPU cross-scene setting. The results demonstrate that all loss terms contribute positively to overall performance, with particularly pronounced effects on balancing known-class recognition and unknown-sample rejection, as reflected by HOS. Specifically, removing the target-domain uncertainty regularization $\mathcal{L}_{ent}$ leads to a substantial degradation, with the average HOS dropping by 17.18% and the average ALL decreasing by 5.33%. This observation indicates that uncertainty regularization in the target domain plays a critical role in stabilizing open-set discrimination under cross-scene shifts, as its absence makes the model prone to incorrect known-class decisions and severely undermines balanced performance. When the negative–negative structural constraint $\mathcal{L}_{nn}$ is removed, the average HOS decreases by 1.89% and the average ALL drops by 3.33%, demonstrating that imposing structured constraints among multiple negative prompts is beneficial for maintaining the effectiveness and stability of negative discriminative semantic cues. Furthermore, removing the positive–negative structural constraint $\mathcal{L}_{pn}$ results in a 1.58% decrease in average ALL, suggesting that explicitly modeling the relationship between positive and negative semantics further reinforces category-level decision boundaries.

2) Feature Distribution: Fig. 5 shows t-SNE visualizations of different OSDA methods on the target domain, where

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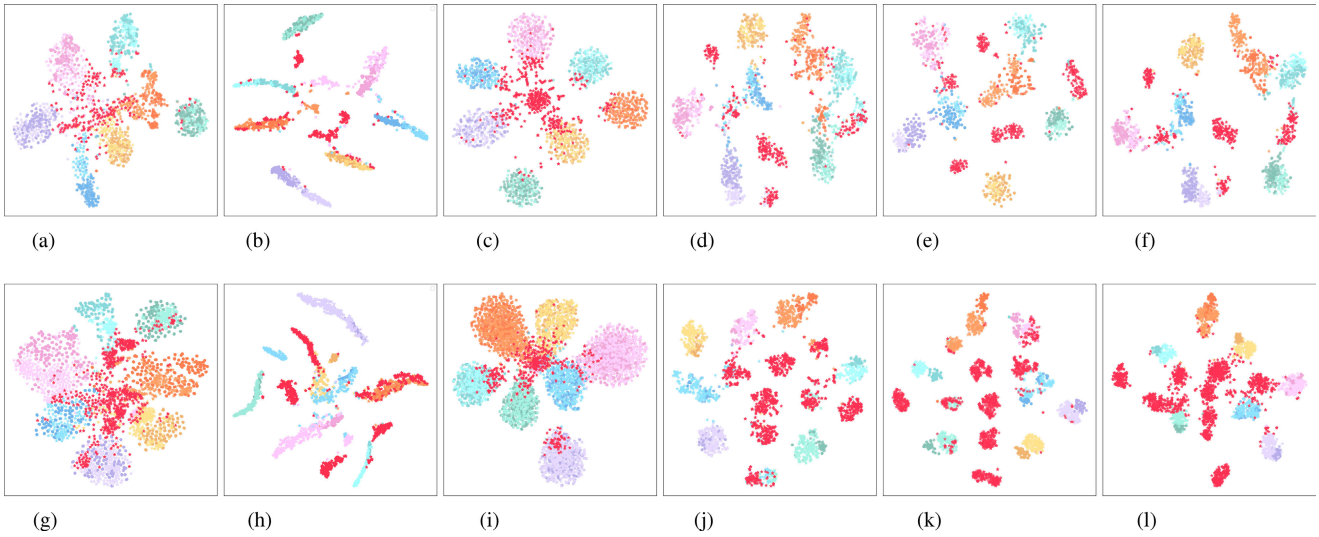


Fig. 5. T-SNE visualizations of feature distributions learned by different methods in the AID_CLRS_MLRSN_OPT setting. Each color denotes a distinct category. Features from the source and target domains are marked by circles (o) and asterisks (*), respectively. (a) DAMC M→O. (b) PUMCL M→O. (c) OSBP M→O. (d) OSClip M→O. (e) TASC M→O. (f) OURS M→O. (g) DAMC O→A. (h) PUMCL O→A. (i) OSBP O→A. (j) OSClip O→A. (k) TASC O→A. (l) OURS O→A.

TABLE V
ABLATION STUDY ON SINGLE LOSS COMPONENT REMOVAL (AID_UCM_NWPU)

Component	Metric/Task	A→U	A→N	U→A	U→N	N→A	N→U	Avg.
w/o $\mathcal{L}_{ent}$	HOS	77.01	65.59	78.11	79.90	74.20	72.61	74.57
	ALL	83.25	78.20	**86.79**	80.70	90.08	85.00	84.00
w/o $\mathcal{L}_{nn}$	HOS	91.34	87.73	**90.22**	82.73	93.19	93.93	89.86
	ALL	89.33	89.66	85.73	65.75	94.12	91.42	86.00
w/o $\mathcal{L}_{np}$	HOS	91.63	88.87	86.99	91.51	**93.98**	92.78	90.96
	ALL	88.92	**90.12**	78.15	84.45	94.50	90.33	87.75
OURS	HOS	**91.90**	**88.98**	89.98	**92.43**	93.00	**94.18**	**91.75**
	ALL	**89.50**	89.19	85.05	**85.42**	**95.07**	**91.75**	**89.33**

different colors denote known categories and red points indicate unknown samples. Traditional OSDA methods exhibit loosely clustered known categories with ambiguous class boundaries, while VLM-based approaches improve known-class compactness but still place unknown samples close to multiple known clusters. In contrast, the proposed method forms more compact and well-separated known-class clusters and consistently pushes unknown samples away from all known categories. These observations suggest that the proposed PNKUS strategy facilitates more stable known–unknown separation by contrasting positive and negative category-level binary semantics under cross-scene conditions.

3) Unknown Categories: As shown in Fig. 6, as the number of unknown categories in the target domain increases, the proposed method exhibits a consistent upward trend in both HOS and ALL on the C→A and M→C transfer tasks, in contrast to most competing approaches. This behavior indicates that the proposed method establishes stable positive–negative semantic discrimination under increasingly open label spaces, rather than treating unknown categories as mere interference. As the number of unknown categories grows, the joint use of multiple negative prompts and positive–negative semantic differences enables more effective modeling of unknown-class

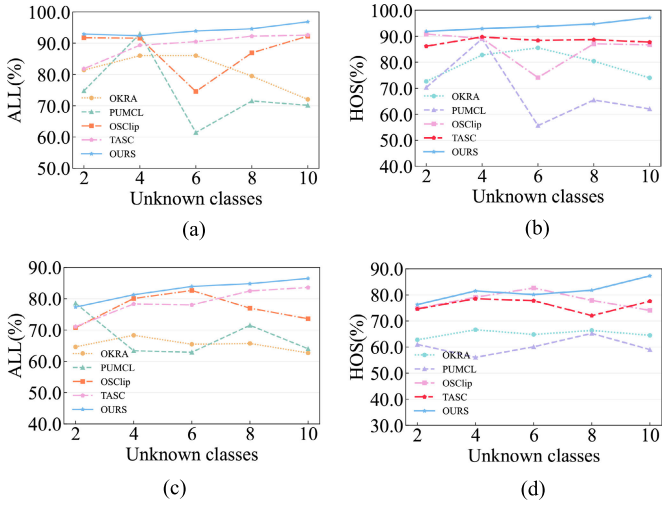


Fig. 6. HOS (%) and ALL (%) of different methods under varying numbers of target-domain unknown categories across multiple transfer tasks in the AID_CLRS_MLRSN_OPT setting. (a) C→A ALL. (b) C→A HOS. (c) M→C ALL. (d) M→C HOS.

semantics, thereby strengthening the boundary between known and unknown categories. In contrast, existing methods often experience performance fluctuations or degradation as the

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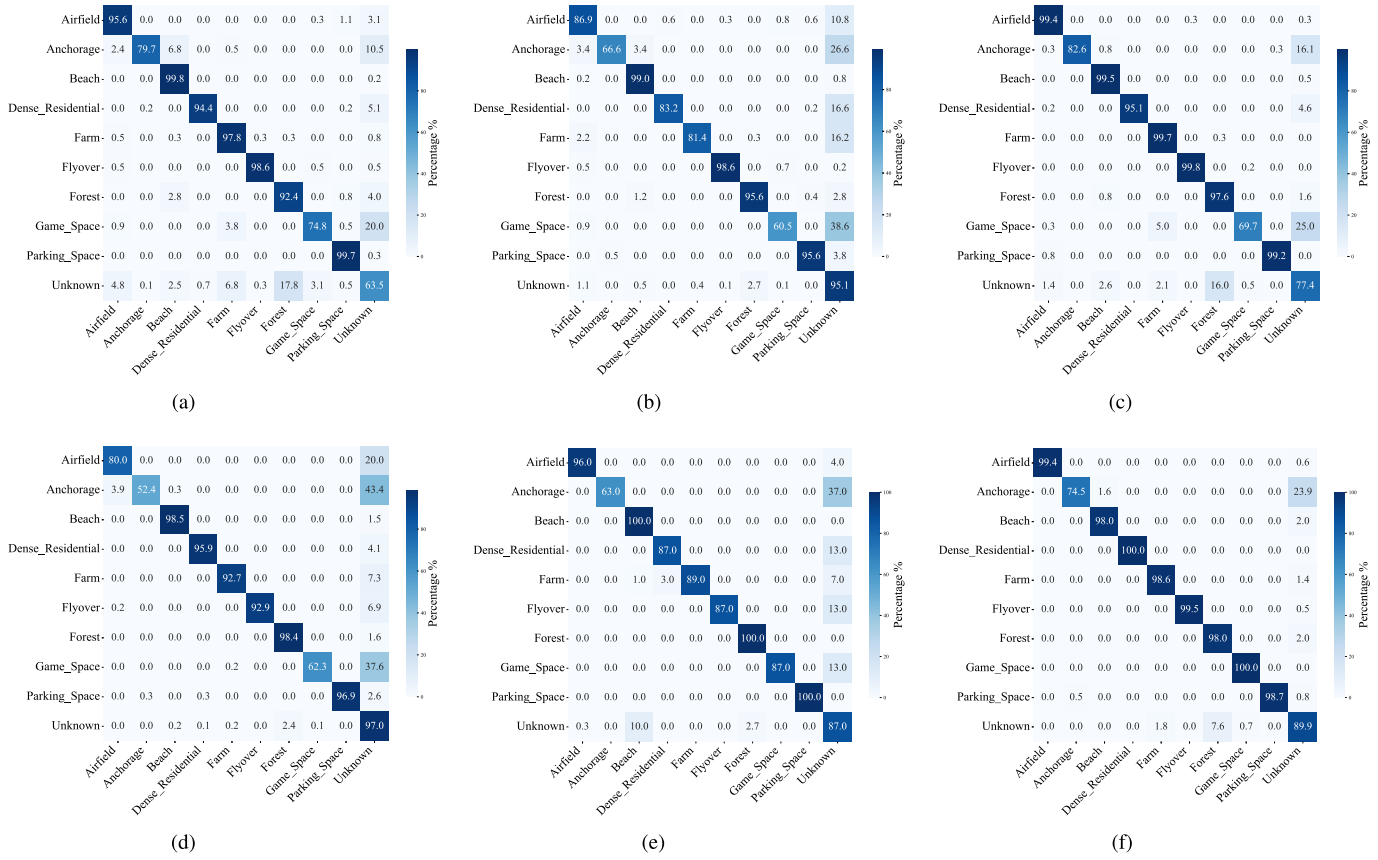


Fig. 7. Confusion matrices of various methods of the target domain on the N→A transfer task under the AID_UCM_NWPU setting. (a) OVANet. (b) UADAL. (c) OKRA. (d) PUMCL. (e) OSClip. (f) OURS.

proportion of unknown categories increases, reflecting their sensitivity to the scale of unknown categories. Overall, these results indicate that the proposed method maintains robust performance under high unknown-category ratios.

4) Confusion Matrices: As shown in Fig. 7, the confusion matrices on the N→A task under the AID_UCM_NWPU setting reveal clear differences among methods. OVANet [47] and UADAL [12] suffer from noticeable confusion among known categories and weak rejection of unknown samples, while OKRA [34] and PUMCL [17] improve known-class recognition but still disperse unknown samples across multiple known categories. OSClip [21] achieves more stable known-class predictions, yet still exhibits nonnegligible acceptance of unknown samples. In contrast, the proposed method yields higher accuracy on known categories and consistently assigns most unknown samples to the unknown label, demonstrating a clearer and more reliable known–unknown separation under cross-scene conditions.

5) Sensitivity Analysis of Parameters: Our method involves five key hyperparameters, including the number of negative prompt groups M, the prompt depth J, the prompt length n, and the loss weights α and β.

Table VI reports the averaged results over 12 transfer tasks in the AID_CLRS_MLRSN_OPT setting to analyze the impact of the number of negative prompt groups, with $M \in \{1, 2, 3, 4, 5, 6\}$. The results indicate that the number of negative prompt groups has a noticeable effect on model performance.

TABLE VI
ABLATION STUDY ON THE NUMBER OF NEGATIVE PROMPT GROUPS M (AID_CLRS_MLRSN_OPT)

Metric/Task	M = 1	M = 2	M = 3	M = 4	M = 5	M = 6
OS*	87.43	87.55	86.52	**88.50**	87.23	87.20
UNK	89.10	89.31	89.91	90.17	89.65	**90.39**
HOS	88.08	88.29	88.00	**89.21**	88.24	88.61
ALL	88.41	88.58	88.38	**89.47**	88.58	88.99

The strongest overall performance is achieved at $M = 4$, where OS*, HOS, and ALL reach 88.50%, 89.21%, and 89.47%, respectively. This suggests that introducing multiple negative prompt groups to model unknown-class semantics from diverse semantic perspectives helps stabilize open-set discrimination and sharpen decision boundaries. When M is further increased, performance gains tend to saturate, indicating that an excessive number of negative prompts may introduce semantic redundancy or increase optimization difficulty.

As shown in Fig. 8, we analyze the effects of prompt depth $J \in \{1, 3, 6, 9, 12\}$ and prompt length $n \in \{1, 2, 3, 4, 5\}$ in the AID_CLRS_MLRSN_OPT setting. Increasing the prompt depth improves HOS and achieves the best performance at $J = 9$, while deeper configurations lead to saturation or slight degradation. For prompt length, although different tasks exhibit varying sensitivity to n, the setting $n = 2$ provides the most stable overall performance. Based on these observations, $J = 9$ and $n = 2$ are adopted as the default configuration in

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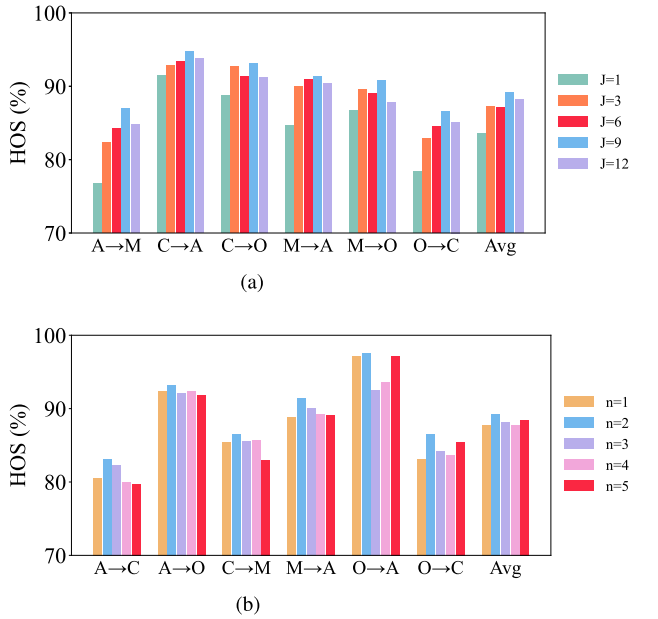


Fig. 8. Ablation results for (a) prompt depth J and (b) prompt length n across six randomly selected transfer tasks under the AID_CLRS_MLRSN_OPT setting, illustrating their impact on model performance.

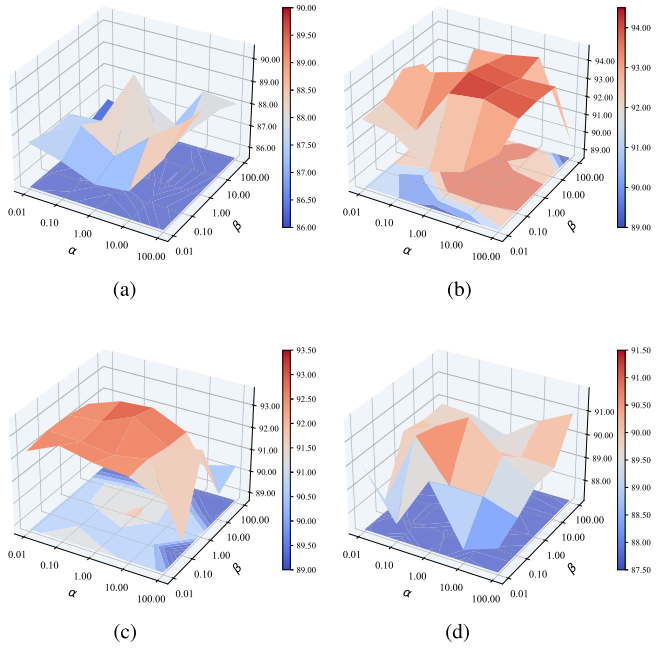


Fig. 9. Hyperparameter sensitivity analysis of α and β on HOS (%). (a) U→A and (b) N→U transfer tasks are selected from the AID_UCM_NWPU setting, while (c) C→O and (d) M→A transfer tasks are selected from the AID_CLRS_MLRSN_OPT setting.

subsequent experiments to balance performance and training stability.

We further analyze the influence of α and β on representative transfer tasks from different cross-scene settings. This is mainly because α and β control two complementary aspects of the framework, i.e., negative-prompt diversity and positive–negative semantic separation, which jointly contribute to robust binary semantic modeling. As shown in Fig. 9,

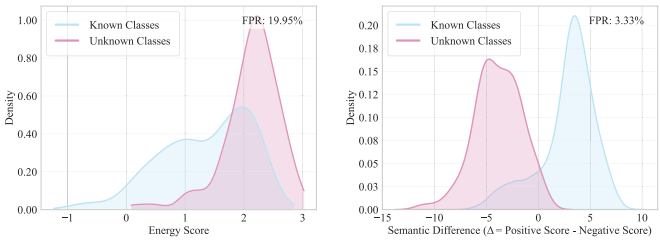


Fig. 10. Density distributions of (a) energy score and (b) semantic difference Δ for known/unknown categories on the A→U transfer task under the AID_UCM_NWPU setting.

TABLE VII
COMPUTATIONAL EFFICIENCY COMPARISON OF DIFFERENT METHODS

Metric/ Methods	Trainable Params (M)↓	Testing Time (s/batch)↓	Training Time (s/batch)↓
OSBP	27.445	**0.0012**	**0.0042**
DAMC	30.658	0.0126	0.0477
UADAL	27.380	0.0058	0.0538
OVANet	23.563	0.0168	0.0271
SSOUDA	26.085	0.0124	0.1417
MAOSDAN	27.574	0.0116	0.1162
OKRA	24.038	0.0167	0.0603
PUMCL	26.154	0.0133	0.4586
OSClip	0.898	0.1051	0.1602
TASC	**0.492**	0.0813	0.1775
OURS	17.775	0.0442	0.1700

HOS (%) varies with $\alpha, \beta \in \{0.01, 0.10, 1.00, 10.00, 100.00\}$, indicating that the two weighting factors exhibit moderate task-dependent sensitivities under different cross-scene discrepancies. In particular, larger α values tend to benefit scenarios with stronger domain discrepancies or more complex unknown-category interference by enhancing negative semantic diversity, whereas larger β values can improve positive–negative semantic discrimination when sharper known–unknown separation is required. Overall, the performance remains stable within a relatively wide parameter range, suggesting that CLPNP does not rely on highly sensitive parameter tuning. Setting both parameters to 1.0 achieves a favorable balance between semantic diversity and discrimination strength, leading to stable and competitive performance across different transfer tasks.

6) Density Distributions: Fig. 10 compares the density distributions of the energy score and the proposed semantic difference metric Δ for known and unknown categories. As shown in Fig. 10(a), the energy score distribution of OSClip [21] exhibits substantial overlap between known and unknown categories, leading to a relatively high FPR of 19.95%. In contrast, Fig. 10(b) presents the distribution of the proposed semantic difference metric Δ ($\Delta = s^{+}(x) - \bar{s}(x)$), where known and unknown categories are more clearly separated, and the FPR is significantly reduced to 3.33%. These results demonstrate that the proposed semantic difference metric enables more reliable known–unknown discrimination and effectively reduces misclassification in OSDA tasks.

7) Computational Efficiency Analysis: To provide a clearer evaluation of computational efficiency, Table VII reports the trainable parameters, testing time, and training time of

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different methods under the same implementation setting. All experiments are conducted on a workstation equipped with an NVIDIA GeForce RTX 4090 GPU, and the time cost is reported in seconds per batch. As shown in Table VII, CLPNP has 17.775 M trainable parameters, with a testing time of 0.0442 s/batch and a training time of 0.1700 s/batch. It should be noted that the increased trainable parameter count mainly stems from the category-level prompt design, where each known category is associated with a dedicated positive prompt and multiple groups of dedicated negative prompts to explicitly model category-level belonging and nonbelonging semantics, while the vision–language backbone remains frozen. Despite this increase, CLPNP still achieves faster inference than lightweight VLM-based baselines such as OSClip [21] and TASC [48], while maintaining a comparable training time.

E. Discussion of Limitations

Although the proposed method achieves promising performance in open-set cross-scene RS image classification, it still has certain limitations. To explicitly model category-level belonging and nonbelonging semantics, the proposed framework assigns each known category a dedicated positive prompt together with multiple groups of dedicated negative prompts. While this category-specific positive–negative prompt construction enhances open-set discriminability, it also leads to a relatively larger number of trainable parameters. Future work will focus on developing more lightweight and adaptive prompt modeling strategies, such that the prompt structure can be adjusted according to category semantic complexity and cross-scene discrepancy, thereby improving parameter efficiency and flexibility while preserving strong open-set discrimination capability.

V. Conclusion

This article presents CLPNP, a category-level positive and multinegative prompt learning framework for open-set cross-scene RS image classification. By explicitly introducing both class-belonging and class-nonbelonging semantics into a unified vision–language representation, CLPNP establishes a category-level binary semantic discrimination mechanism and goes beyond existing approaches that rely solely on positive semantic matching. Built upon this prompt design, the proposed PNKUS strategy explicitly shapes binary semantic boundaries in the source domain through positive–negative semantic contrast, while suppressing the semantic ambiguity of unlabeled target samples through uncertainty regularization. Moreover, a threshold-free inference mechanism enables open-set prediction by directly comparing positive and negative semantic responses for each known category. Extensive experimental results across multiple open-set cross-scene transfer settings demonstrate that CLPNP achieves consistently superior and well-balanced performance in known-category recognition and unknown-sample rejection. In future work, we will explore extending positive–negative semantic prompting to more challenging open-set settings (e.g., universal domain adaptation) and a wider range of RS tasks.

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